AGI ALPHA: A Recoverable Enterprise That Finances Productive Expansion

Montreal.AI / AGI ALPHA engineering campaign

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Abstract

The registered design completed 8 separate simulated enterprises. In the central growth region, 4/4 met the predeclared commercial criteria. Independent checks reexecuted useful artifacts and replayed native financial records and operating semantics. These finite authored trajectories are not calibrated probabilities of real market success.

Source SHA256: `3d985299bf6cfaf546b712ebd9e878e8 75aaf730ffdb32023637aafc5504a38d` **Committed source:** `ffc1f2e14e88e11f75d5087ab0f7c4e60512431e`

1 Achieved result

The registered design completed 8 separate simulated enterprises. In the central growth region, 4/4 met the predeclared commercial criteria. Independent checks reexecuted useful artifacts and replayed native financial records and operating semantics. These finite authored trajectories are not calibrated probabilities of real market success.

2 Representative enterprise

The first registered central growth enterprise collected $898.225750 from 28 settled complete deliveries to 11 paying customers, including 10 repeat customers. Complete modeled expenses were $361.241043; profit and available earned funds were $536.984707 and $536.984707. 23 profitable deliveries used prior unrestricted earnings. Initial seed capital was $5,000.000000; terminal seed-backed protection was $1,003.825750.

3 Independent registrations and failure preservation

The main expansion addendum registered four fresh paired seeds and a new cohort before execution, over unchanged qualified business source002. It compares owned inbound operation with retained financed acquisition and capacity, using separate explicit USD 5,000 seed allocations. No earned balance is copied. Source001 previously preserved 15 results and 14 accepted operating audits before a genuine exposure/reactivation trace-order failure; its incomplete verdict was never promoted. Source002 corrected the trace and used new seeds. The local clean-package route also preserved one unexplained missing-case-field rejection; a fresh unchanged retry and an independent hosted route passed.

4 Matched growth consequence

Under matched starting capital and contracts, central growth minus baseline profit ranged from $297.813404 to $692.558932, with mean $445.914756. This is the value of the added commercial options and resulting behavior,

not proof that a more elaborate algorithm is superior.

5 Preserved comparison and policy selection

The separate 20-business pricing study completed all independent audits. Its pricing option met central commercial criteria in 4/4 cases, but only 1/4 central pairs improved both delivery volume and profit, below the registered 3/4 criterion. Central mean profit changed by $-5.731327. Retained ordinary growth was therefore preserved as the central method. In the two pressure cases the option collected 16 and 22 deliveries versus 0 for retained operation, with positive incremental profit. Regional utility does not establish general algorithmic superiority. The full pricing-study report and every failure remain published separately.

6 Authenticated recovery and intervention

One addendum account stopped because its native tape contained 246 actions while its intact signed command journal and sealed state bound 397. The native account failed closed. A separately published, bounded recovery phase used a shadow dry run and canonical owned Account calls to restore only the 151 original authenticated commands. All 367 stored receipt-prefix checks and the sealed 397-action digest matched; keys, journal, operating state and obligation proofs remained unchanged. The original accounts then completed their saved queues. This was AI-operator-assisted recovery, not autonomous self-repair or proof of hostile-host security. The original runtime interruption and monitoring gaps are preserved; no receipt, seed capital or economic outcome was fabricated.

7 Connected operating path

Voluntary budget-feasible qualified order precedes production. The admitted service executor produces the actual artifact; a separate checker verifies source-bound complete outcomes before native prepare. Hour-clock clearing follows acceptance. Paid acquisition and capacity contracts are bought only from current unrestricted earnings using an audited state-bound reservation and resolved prepaid resource obligations. Expiring channels expose finite prospects with overlap, fatigue, failed access and renewal costs; exposure is never an order. Customers retain human, rival and refusal alternatives even when the enterprise lacks capacity.

8 Recovery and independent verification

Every monetary command is recorded as an exact signed write-ahead intent before native mutation. Operating state, event queue and prepared decision are separately sealed against the retained local simulation key and a native tape prefix. Crash tests cover startup, qualified order, production, acceptance and clearing/tax acknowledgment loss. Public exports omit private keys and cannot reopen accounts; they support independent audit and fresh reproduction. Storage protection is limited to the supported operator-owned local POSIX filesystem.

9 Cost and inference limits

All amounts are simulated microUSD. Complete modeled costs include development/search/qualification/setup, acquisition including unsuccessful and unused exposure, onboarding, production, supplier, verification/support, correction envelopes, maintenance, payment fees and conditional tax. Exact pinned runtime dependencies and local CPU are checked or recorded separately. Host and model billing are unavailable and remain explicit envelopes or unknown; no zero-billing inference is made. Human complete-work hours and fully loaded wage proxies are authored commercial references. No actual paid frontier-model inference, live customers, outreach, financial transactions or deployment occurred.

10 Scale and unmet milestones

Exact aggregation of native records reconciles overlapping collected-delivery prefixes and all expense/cash flows. This validates accounting aggregation, not a forward buyer or production model. USD1M,10M,100M,1B and10B trailing-365-day targets remain unreached in this finite native experiment. The separate PR22 months22-23 and PR23 months27-29 hypothetical billion-dollar projections use different capital, offers, channel contracts, allocation policies, outreach costs and reserve semantics. They remain UNBRIDGED_NATIVE_SCALE; neither number is selected as the native enterprise forecast.

11 Boundaries and next action

The operating implementation, conditional commercial result and growth result are separate outcomes. Small synthetic complete services and authored buyer/channel/supplier behavior do not establish empirical market demand or autonomous invention superiority. The frozen source inventory, finite audience and permanent native refund tails remain real edition constraints. Follow-on construction must earn or explicitly seed its own resources; no balance is copied between evaluation businesses. Use matched realized incremental contribution to retain/renew channels and capacity; close finite artifact/audience/working-capital constraints with a new costed qualified capability before claiming a native scale bridge. Historical aggregate billion projections remain separate hypotheses.

12 Unpriced resource sensitivity

Actual host and model billing is unavailable. The representative modeled profit would be eliminated by additional unpriced enterprise costs of $536.984707. The declared setup, service, maintenance and support envelopes are simulation assumptions; they are not zero-cost or measured-billing assertions.

13 All registered economic regions

Region	Arm	N	Paid	Modeled profit range	Covered
Expansion	baseline	4	9-13	$125.197687 to $239.171303	4/4
Expansion	growth	4	27-39	$536.984707 to $817.756619	4/4
Pricing: central	competitive growth	4	31-39	$573.657027 to $765.863824	4/4
Pricing: central	growth	4	29-36	$581.229157 to $727.621272	4/4
Pricing: delay / cost	competitive growth	2	8-10	$78.448560 to $137.825980	2/2
Pricing: delay / cost	growth	2	7-10	$63.334503 to $144.720212	2/2
Pricing: pressure	competitive growth	2	16-22	$142.901645 to $208.297708	2/2
Pricing: pressure	growth	2	0-0	$-89.650000 to $-89.250000	2/2
Pricing: weak reach	competitive growth	2	9-9	$121.901047 to $130.077452	2/2
Pricing: weak reach	growth	2	7-9	$84.441780 to $126.841047	2/2

14 Representative financial state

Measure	Simulated USD
Recognized revenue	$898.225750
Collected payments	$898.225750
Incurred complete modeled costs	$361.241043
Paid costs	$361.241043
Profit	$536.984707
Cash	$5,536.984707
Receivables	$0.000000
Unpaid liabilities	$0.000000
Native tail reserves	$903.825750
Custody reserves	$0.000000
Working reserve	$100.000000
Unrestricted earned funds	$536.984707
Seed capital	$5,000.000000
Seed-backed protection	$1,003.825750

Protected balances overlap: total seed-backed protection includes the native tail and working reserve.

15 Source and reproduction

https://github.com/MontrealAI/AGI-Alpha-Invention-Flywheel-Lab

Read the operator procedure and preserve the registered source, source/configuration manifest, all native financial tapes, accepted artifacts, independent audit receipts and every non-reaching or failed trajectory. The browser edition and Markdown report are generated from the same achieved records.